

Mechanics Domain

LESSON 04
Build The Robotic Arm.

Time Frame: 90 mins

Objectives:

- ☐ Utilize Pascal's Principle to enhance the force application in hydraulic robotic arms.
- ☐ Apply Bernoulli's Principle in the design and operation of pneumatic robotic arms.
- ☐ Prevent or mitigate resonance effects that could impair the robotic arm's performance.

SEL Relation:

Collaboration: Working in groups to build the robotic arm fosters teamwork, communication, and shared responsibility.

Problem-Solving: Overcoming challenges during the construction and testing phases requires perseverance, critical thinking, and the ability to learn from mistakes.

Self-Efficacy: Successfully building a functional robotic arm can boost students' confidence in their abilities

Challenge question for this lesson:

Your team needs to design a robotic arm that can precisely pick up and stack three different irregularly shaped objects. Describe your team's process for brainstorming design ideas, deciding on a final design, and dividing the construction tasks effectively.

The standard gripper on your robotic arm struggles to handle small, flat metallic items. Propose a modification to the gripper's design, including the materials you would use and how it would function, to allow it to reliably pick up and hold these metallic objects.

1. Engage

- **Teacher Role:**
 - Present realistic scenarios involving the need to amplify force in practical situations.
 - Guide students to identify the limitations of direct force application.
 - Introduce the idea of using a fluid-based system as a potential solution.
 - Facilitate student brainstorming and problem-solving related to the scenarios.
- **Expected From Students:**
 - Analyze the presented scenarios and identify the challenges related to force application.
 - Propose initial ideas and potential solutions, even if they are not fully formed.
 - Participate actively in discussions and share their reasoning.
 - Begin to recognize the potential benefits of a system that can multiply force.
- **Parts of the books included:**
 - Likely sections covering the concepts of force, pressure, work, and possibly introductory examples of simple machines. The focus will be on the limitations of applying force directly in certain situations.
- **Intro Scenario:** Use engaging strategies to grab students' attention using realistic scenarios. (List strategies teachers can use to deliver this part, including the name of the strategy and how to implement it.)
 - **Strategy 1: The "Stuck Valve Challenge" (Problem-Based Scenario):**
 - **Implementation:** The teacher presents this scenario: "Imagine you are a robotics technician working on a large industrial robot. One of the hydraulic valves responsible for a powerful arm movement is stuck. You need to apply a very large force to manually override it. You have limited space and your own strength isn't enough. What kind of system or tool could you use to generate the necessary high force with a smaller input force?" The teacher can show a visual of a large valve or a robotic arm joint.
 - **Strategy 2: "Emergency Lift Operation" (Real-World Application Scenario):**
 - **Implementation:** The teacher describes a situation: "A heavy piece of equipment has fallen and is blocking an emergency exit. You need to lift it quickly and safely. You have access to a hydraulic jack and some basic tools. Explain how you could use the hydraulic jack to lift the heavy object, even though you might not be able to lift it directly." The teacher can provide a simple diagram of a jack and the object.
 - **Strategy 3: "Designing a Powerful Gripper" (Engineering Design Scenario):**
 - **Implementation:** The teacher poses a design challenge: "You are tasked with designing a robotic gripper that needs to exert a very strong clamping force to hold heavy or irregularly shaped objects securely. Direct motor actuation might be too bulky or not provide enough force. How could you use a fluid-based system to achieve this high gripping force in a compact design?" The teacher can show examples of different robotic grippers.

- **Scenario Reflection:**

- **Questions:**

- In the "Stuck Valve Challenge," what were the limitations of trying to move the valve with direct human strength? (Focus: Force magnitude)
 - In the "Emergency Lift Operation," how does the hydraulic jack help you lift something heavier than you could normally lift? What's the key difference in how you apply force? (Focus: Force multiplication and application method)
 - In the "Designing a Powerful Gripper," what are the advantages of using a fluid system for generating high force in a robotic gripper compared to just using a motor? (Focus: Force density, potential for force multiplication)
 - What common element seems to be present in all these scenarios that allows a smaller input force to create a larger output force or solve a force-related problem? (Guide towards the idea of a principle at work)

- **Suggested answers:**

- (Stuck Valve): Not enough strength, awkward angle, risk of injury.
 - (Emergency Lift): The jack allows a smaller force applied over a larger distance to create a larger force over a smaller distance.
 - (Powerful Gripper): Potential for higher force in a smaller space, controlled force application.
 - (Common element): Some kind of system that changes or multiplies the applied force, likely involving a fluid.

- **(List strategies teachers can use to deliver this part, including the name of the strategy and how to implement it.)**

- **Strategy 1: Small Group Discussion with Scenario Analysis:**

- **Implementation:** Divide students into small groups and assign each group one of the scenarios to analyze. Have them discuss the challenges, potential solutions, and the role of force in the situation. Each group can then share their key findings with the class.

- **Strategy 2: "Think-Aloud" Problem Solving (Teacher-Led):**

- **Implementation:** The teacher can walk through one of the scenarios, thinking aloud their process of identifying the problem, considering possible approaches, and highlighting the need for a principle that allows force amplification. This models problem-solving for the students.

- **Strategy 3: Brainstorming "Force Multipliers":**

- **Implementation:** After presenting a scenario, ask students to brainstorm different tools or systems they know of that help multiply force (e.g., levers, gears, jacks). This connects the current scenario to their existing knowledge and sets the stage for introducing Pascal's Principle as another such "force multiplier."

- **Strategy 4: "What If?" Scenario Variations:**

- **Implementation:** After discussing an initial scenario, the teacher can introduce variations or complicating factors ("What if the space was even tighter?" or "What if you needed to lift it much

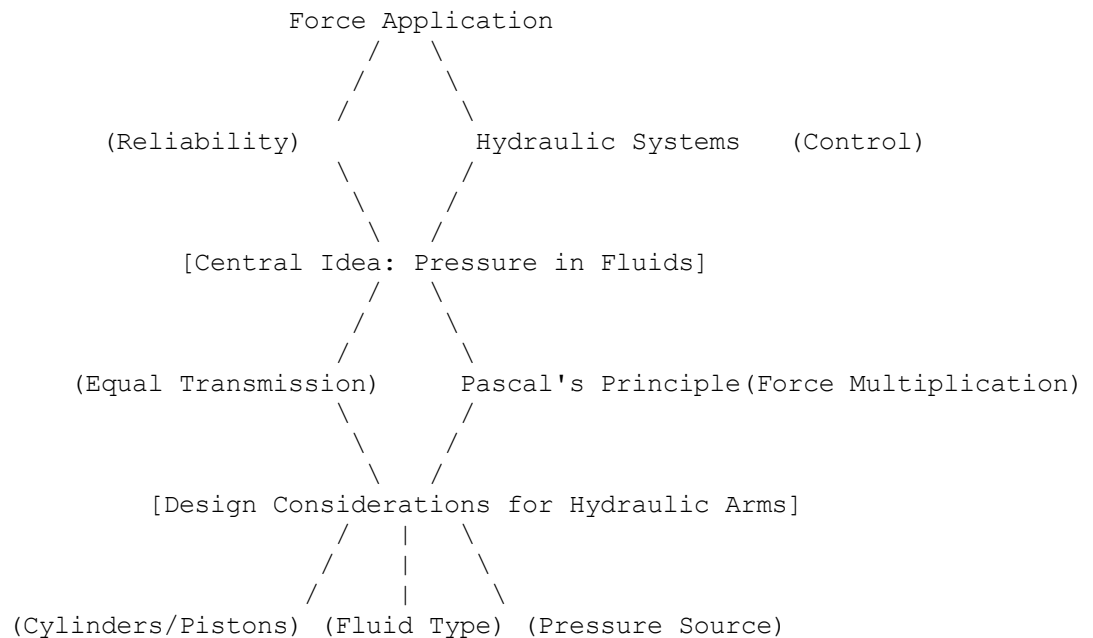
higher?"). This encourages deeper thinking about the limitations of different approaches and the need for a robust principle like Pascal's.

2.Think

- **Questions and Possible Answers:**

- o **Question 1 :**What type of pressure works best for your robotic arm?
 - **Possible Answers (Leading towards Pascal's Principle):**
 - Fluid pressure (hydraulic pressure) because it can be transmitted equally in all directions.
 - High pressure to generate large forces.
 - Consistent and reliable pressure for controlled movements.
 - Pressure that can be easily controlled and adjusted.
 - *Teacher Note:* Guide students to connect "best" to the idea of efficient force transmission and amplification, which is central to Pascal's Principle.
- o **Question 2 :** How can you design your robotic arm to stay stable and avoid unwanted movements?
 - **Possible Answers (While not directly Pascal's Principle, it's important for overall design):**
 - Using a stable base and strong joints.
 - Employing counterweights or balancing mechanisms.
 - Designing a rigid structure.
 - Implementing control systems with feedback loops.
 - Carefully considering the distribution of mass.
- o **Additional Question (Connecting to Pascal's Principle):** How could you use the properties of fluids and pressure to make your robotic arm exert a much larger force than the initial force you apply?
 - **Possible Answers:**
 - By using a smaller cylinder to apply pressure to a larger cylinder, the force will be multiplied.
 - By transmitting pressure through a confined fluid.
 - By understanding that pressure applied to a fluid is transmitted equally throughout the fluid.

Brainstorm Web:



- (List strategies teachers can use to deliver this part, including the name of the strategy and how to implement it.)
 - **Strategy 1: Individual Brainstorming followed by Group Sharing (Think-Pair-Share Variation):**
 - **Implementation:** First, pose the "Think" questions to the students and give them a few minutes to individually jot down their initial ideas and answers. Then, have them pair up with a partner to discuss their thoughts and build upon each other's ideas. Finally, facilitate a whole-class discussion where pairs share their key insights.
 - **Strategy 2: Guided Questioning with Wait Time:**
 - **Implementation:** Pose the "Think" questions one at a time, allowing for sufficient "wait time" after each question. This encourages more students to formulate their own answers instead of immediately relying on others. You can then solicit responses from a diverse range of students.
 - **Strategy 3: Concept Mapping (Visual Organization):**
 - **Implementation:** Introduce the central concept (e.g., "Pressure in Hydraulic Arms") and guide students to create a concept map, branching out with related ideas, properties, and design considerations. This helps them visually organize their thinking and see the connections between different concepts.
 - **Strategy 4: "What If?" Scenarios for Deeper Thinking:**
 - **Implementation:** Present "what if" scenarios related to the questions. For example, "What if the pressure wasn't transmitted equally?" or "What if

you used air instead of oil?" These scenarios can challenge their initial thinking and push them to consider the underlying principles more deeply.

- **Strategy 5: Using Analogies to Bridge Understanding:**

- **Implementation:** Relate the concept of pressure transmission to everyday experiences. For example, "Think about squeezing a toothpaste tube – the pressure you apply at one end is transmitted throughout the toothpaste." This can help students grasp the abstract concept of pressure.

3.Explore

- **Teacher Role:**

- Provide clear and concise instructions for each activity.
- Ensure students have access to the necessary materials and use them safely.
- Circulate among student groups, observing their progress and providing guidance as needed.
- Ask probing questions to encourage deeper thinking and connection to the learning objectives.
- Facilitate discussions and ensure all students have an opportunity to share their observations and insights.
- Help students connect their experimental findings to the principles of Pascal's Principle, Bernoulli's Principle, and resonance.

- **Expected From Students:**

- Actively participate in each hands-on activity.
- Follow the step-by-step instructions carefully.
- Make careful observations and record their findings.
- Work collaboratively within their groups (if applicable).
- Discuss their observations and try to answer the reflection questions based on their experimental evidence.
- Ask clarifying questions when needed.
- Begin to connect their observations to the behavior and design considerations of robotic arms.

- **Parts of the books included:**

- **Pascal's Principle:** Concepts of pressure, force transmission in fluids, hydraulic systems.

Connect two syringes of different sizes with tubing and fill them with water. Push on the smaller syringe and observe the movement of the larger one.

Observation Questions:

- What happens to the larger syringe when you push the smaller one? How does the force change?
- How does changing the size of the syringes affect the force and movement?
- How could this principle be used to design a robotic arm that can lift heavy objects with minimal effort?

- **Bernoulli's Principle:** Concepts of fluid flow, pressure differences, lift and drag.

Use the hairdryer to blow air over the top of a ping pong ball. Observe how the ball behaves in the airflow. Then, create and fly paper planes with different designs.

Observation Questions:

- What happens to the ping pong ball when air blows over it? Why does it stay in the air?
- How does the design of the paper plane affect its flight? Which design flies the furthest or stays in the air the longest?
- How can controlling airflow in a robotic arm's pneumatic system make it move more smoothly?

- **Resonance:** Concepts of vibration, natural frequency, and the effects of matching frequencies.

Explore





Attach small weights to the end of a flexible ruler or rod. Strike the tuning fork and hold it near the ruler to observe the vibration.

Observation Questions:

- What happens to the ruler when the tuning fork vibrates at different frequencies?

- How does the amount of weight on the ruler affect its vibration?

- What does this suggest about how a robotic arm might react to different forces during operation?

- **Activity - Do It:**

- **Activity 1 (Related to Pascal's Principle):**

- **Materials Needed:** Two syringes of different sizes, flexible tubing that fits the syringe nozzles, water, a container to fill the syringes.
- **Step-by-Step Instructions:**
 1. Connect the nozzles of the two syringes with the flexible tubing.
 2. Submerge the connected syringes and tubing in water to fill them completely, ensuring no air bubbles are trapped.
 3. Hold one syringe steady and push the plunger of the other syringe inward.
 4. Observe what happens to the plunger of the second syringe.
 5. Try pushing the plunger of each syringe with approximately the same effort and compare the movement of the other plunger.

- **Reflection Questions:**

- What happens to the larger syringe when you push the smaller one? How does the force seem to change?
- How does changing the size of the syringes affect the distance the plungers move and the force exerted?
- How could this principle be used in a robotic arm to lift heavy objects with minimal effort?

- **Activity 2 (Related to Bernoulli's Principle):**

- **Materials Needed:** Hairdryer, ping pong ball, various designs of paper planes (prepared by students or provided).

- **Step-by-Step Instructions:**

- Turn on the hairdryer and direct the airflow upwards at a slight angle.
- Carefully place the ping pong ball into the stream of air. Observe its behavior. Try tilting the hairdryer.
- Fly each of the paper plane designs, observing how their shape affects their flight distance and stability.

- **Reflection Questions:**

- What happens to the ping pong ball when air blows over it? Why does it seem to stay suspended in the air?
- How does the design of the paper plane affect its flight? Which design flies the furthest or stays in the air the longest?
- How can controlling airflow in a robotic arm's pneumatic system make its movements smoother and more controlled?

- **Activity 3 (Related to Resonance):**

- **Materials Needed:** Flexible ruler or rod (e.g., a thin metal strip or a plastic ruler), small weights (washers, paperclips), a tuning fork (ideally with different frequencies, if available).

- **Step-by-Step Instructions:**
 - Securely attach a small weight to the end of the flexible ruler or rod, allowing the other end to be held or clamped.
 - Strike the tuning fork to make it vibrate.
 - Hold the vibrating tuning fork near the weighted end of the ruler/rod, without touching it. Observe the ruler's response.
 - If available, repeat steps 2-3 with tuning forks of different frequencies.
 - Try changing the amount of weight attached to the ruler and repeat the observation with one of the tuning forks.

- **Reflection Questions:**
 - What happens to the ruler when the tuning fork vibrates at different frequencies? Is the vibration of the ruler always the same?
 - How does the amount of weight on the ruler seem to affect its vibration in response to the tuning fork?
 - What does this suggest about how a robotic arm might react to different forces or vibrations during its operation, and why it's important to consider its design to avoid unwanted vibrations?

4. Explain

- **Teacher Role:**

- Explain the concepts of Pascal's Principle, Bernoulli's Principle, and Resonance clearly and concisely, linking them to the "Explore" activities.
 - Guide class discussions to help students connect their observations to the scientific principles and their applications in robotic arms.
 - Address any misconceptions and provide further clarification as needed.
 - Emphasize the rules provided for each principle and their importance in engineering design.
 - Facilitate the "Watch It" and "Read About It" sections to enhance understanding through multimedia resources.
- **Expected From Students:**
 - Engage actively in discussions, sharing their observations and interpretations from the "Explore" phase.
 - Listen attentively to the teacher's explanations and the insights shared by their peers.
 - Ask thoughtful questions to clarify their understanding of the concepts.
 - Make connections between the scientific principles and the functioning of robotic arms.
 - Participate actively in the pre-, during-, and post-watching/reading activities.
 - Summarize the key information learned from the videos and readings.
 - **Parts of the books included:**



- **Watch It:**
 - **Pre-Watching: Set a purpose.**
 - **Teacher Role:** "Before we watch this video about how a robotic arm works, let's think about what we've already explored. We saw how pressure in fluids can be used to create force (Pascal's Principle), how airflow can affect movement (Bernoulli's Principle), and how vibrations can occur (Resonance). As you watch, pay attention to how these ideas might be applied in the design and operation of the

robotic arm shown. What questions do you hope this video will answer for you?"

- **Expected From Students:** Consider their prior experiences from the "Explore" activities and formulate questions they want the video to address regarding the robotic arm's movement, power source, and stability.
- **During Watching: Active engagement.**
 - **Teacher Role:** "As you watch, keep your observations from our experiments in mind. Look for examples of how fluids might be used to generate movement or force. Notice the different parts of the robotic arm and how they might be designed to be stable and avoid unwanted vibrations. You can jot down any key observations or questions that come to mind."
 - **Expected From Students:** Watch the video attentively, focusing on identifying connections to Pascal's Principle, Bernoulli's Principle (if applicable to the arm's design), and considerations for stability related to resonance. They may take notes on what they see and hear.
- **Post-Watching: Discussion and reflection.**
 - **Teacher Role:** "Now that we've watched the video, let's discuss what we saw. Did the video help answer the questions we had beforehand? Can you identify any parts of the robotic arm's operation that seem to rely on Pascal's Principle or Bernoulli's Principle? What design features might be in place to prevent unwanted vibrations or resonance? How does what we saw in the video relate to our experiments with the syringes, ping pong ball, and ruler?"
 - **Expected From Students:** Share their observations and insights from the video, making explicit connections to the scientific principles discussed and their own experimental findings. They should participate in a class discussion, building on each other's ideas.
 - **(List strategies teachers can use to deliver this part, including the name of the strategy and how to implement it.)**
 - **Strategy 1: Think-Pair-Share with Video Focus:** Pose specific questions about the video content (related to the principles) and have students think individually, discuss with a partner, and then share with the class.
 - **Strategy 2: Video Observation Chart:** Provide students with a simple chart to fill out during the video, prompting them to

look for evidence of Pascal's Principle, Bernoulli's Principle, and considerations for resonance in the robotic arm's design.

- **Strategy 3: "Two Stars and a Wish":** After the video, have students identify two things they learned that were particularly interesting or helpful (the "stars") and one question they still have or something they wish the video had explained further (the "wish").
- **Read About It:**
 - **Pre-Reading: Discussion.**
 - **Teacher Role:** "This link will take you to information about the parts of a robotic arm. Before you read, let's brainstorm the different parts you think a robotic arm might have based on the video we just watched and our understanding of how it might move and operate using hydraulic or pneumatic systems. What kind of joints, power sources, and structural elements might be involved?"
 - **Expected From Students:** Share their prior knowledge and predictions about the components of a robotic arm, drawing on their understanding of the underlying principles and their observations from the video.
 - **During Reading: Engagement.**
 - **Teacher Role:** "As you read about the parts of the robotic arm, pay attention to how these parts relate to the principles we've been discussing. Can you identify components that would be part of a hydraulic or pneumatic system? Are there descriptions of how the arm is designed for stability and controlled movement? Highlight or note down any connections you find."
 - **Expected From Students:** Actively read the provided material, looking for specific components and their functions, and making connections to Pascal's Principle (for hydraulic systems), Bernoulli's Principle (for pneumatic systems, if mentioned), and the concept of resonance (in terms of structural stability).
 - **Post-Reading: Summarize and reflect.**
 - **Teacher Role:** "Now that you've read about the parts of the robotic arm, can you summarize the key components and how they contribute to its overall function? How do these parts relate back to Pascal's Principle, Bernoulli's Principle, and the need to avoid resonance? Were there any parts or functions that you found particularly interesting or that helped clarify how these principles are applied in a real-world robotic system?"
 - **Expected From Students:** Summarize the main parts of the robotic arm they learned about and explain how these components utilize or are affected by Pascal's Principle, Bernoulli's Principle, and the

considerations for resonance. They should reflect on how the reading enhanced their understanding.

- **(List strategies teachers can use to deliver this part, including the name of the strategy and how to implement it.)**
 - **Strategy 1: Jigsaw Reading:** Divide the reading material into different sections and assign each section to a small group of students. Each group becomes an "expert" on their section and then shares their key findings with the rest of the class.
 - **Strategy 2: Graphic Organizer:** Provide students with a graphic organizer (e.g., a concept map or a table) to fill in as they read, prompting them to identify the parts of the robotic arm and link them to the relevant scientific principles.
 - **Strategy 3: "Think-Ink-Pair-Share" for Summarization:** After reading, have students first think individually about the key takeaways, then jot down their summary (ink), pair up to discuss and refine their summaries, and finally share their group's summary with the class.

Explanation of Resonance

- **Definition:** Resonance occurs when an object vibrates strongly at its **natural frequency** due to matching **external vibrations**.
- **Rule:** We need to **avoid matching the natural frequency of parts to prevent damage**.

Let's break this down further and relate it to the robotic arm context:

1. **Natural Frequency:** Every object has a natural frequency (or multiple natural frequencies) at which it tends to vibrate when disturbed. Think of plucking a guitar string – it vibrates at a specific pitch, its natural frequency. The natural frequency depends on factors like the object's material, shape, size, and mass distribution. In a robotic arm, different parts like the links, joints, and even the entire structure will have their own natural frequencies.

2. **External Vibrations:** These are vibrations coming from outside the object or system. In the case of a robotic arm, external vibrations could be caused by:
 - The motors and actuators within the arm itself as they move and generate forces.
 - Vibrations from the environment where the robot is operating (e.g., from nearby machinery or the surface it's mounted on).
 - Forces applied to the arm during tasks, especially if these forces have a periodic or repeating nature.
3. **Matching Frequencies (The Danger Zone):** Resonance happens when the frequency of an external vibration is the same as or very close to the natural frequency of a part of the robotic arm. When this match occurs, the energy from the external vibration is transferred very efficiently to the vibrating part. This leads to a dramatic increase in the amplitude (size) of the vibration.
4. **Why Avoid It? (Preventing Damage):** These large vibrations caused by resonance can be very damaging to the robotic arm:
 - **Increased Stress and Strain:** The excessive shaking can put immense stress on the materials of the arm, especially at joints and connection points. This can lead to fatigue, cracks, and eventual failure of components.
 - **Reduced Accuracy and Control:** Large, uncontrolled vibrations make it very difficult for the robotic arm to perform precise movements and tasks accurately. The end effector (the "hand" of the robot) might wobble or overshoot its intended position.
 - **Noise and Wear:** Resonance can generate unwanted noise and increase wear and tear on the moving parts of the arm due to the increased friction and impacts from the amplified vibrations.
 - **Instability:** In extreme cases, resonance can even lead to instability of the entire robotic arm structure.

How This Relates to Robotic Arm Design (The "Rule"):

Engineers designing robotic arms must carefully consider the natural frequencies of all the arm's components and the potential sources of external vibrations during operation. To avoid resonance, they might employ several strategies:

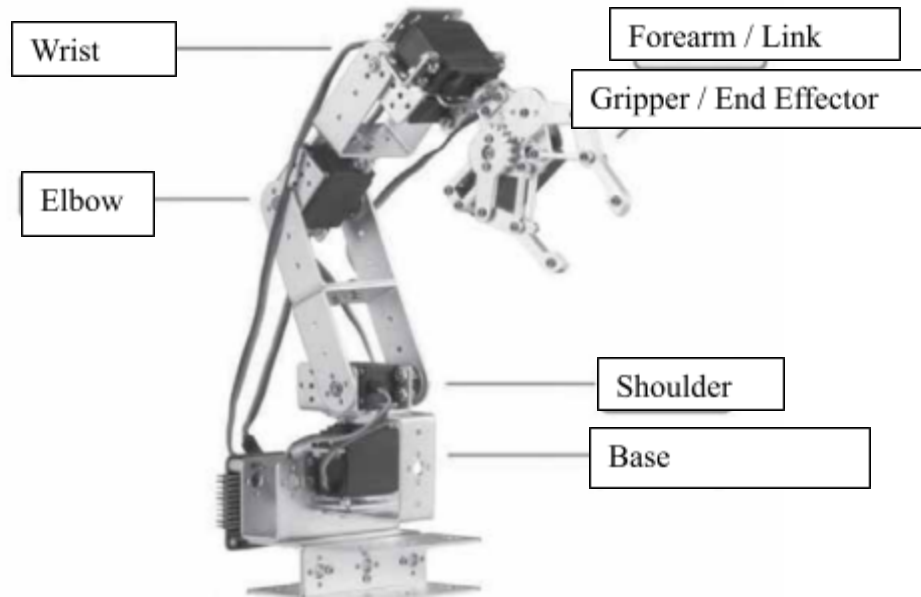
- **Material Selection:** Choosing materials with appropriate damping properties can help dissipate vibrational energy.
- **Shape and Structure Optimization:** Designing the arm's geometry and using structural elements that stiffen the arm can shift its natural frequencies away from expected operating frequencies.
- **Damping Mechanisms:** Incorporating dampers (like shock absorbers) into the joints and structure can help absorb and reduce vibrations.
- **Control Systems:** Implementing sophisticated control algorithms that can actively counteract vibrations or avoid operating at frequencies close to the arm's natural frequencies.
- **Isolation:** Isolating the robotic arm from external sources of vibration through specialized mounts or platforms.

5. Elaborate & Evaluate

- **Teacher Role:**
 - Encourage students to apply their understanding of Pascal's Principle to design and construct a functional hydraulic robotic arm.
 - Foster creativity in their choice of materials and design solutions for the "Practice" activity.
 - Provide guidance and support during the construction process, ensuring safe and effective use of materials.
 - Encourage problem-solving and iterative design as students encounter challenges during the "Practice."
 - Observe and assess students' robotic arm constructions based on functionality, design, and application of Pascal's Principle.
 - Evaluate students' responses to the reflection questions in the "Practice," looking for evidence of understanding and critical thinking.
 - Introduce the "Challenge" as an extension activity encouraging further application of knowledge and creative problem-solving.
 - Assess students' proposed solutions to the "Challenge" based on their feasibility and creativity (if collected).

- Provide constructive feedback to students on their projects and their understanding of the underlying principles throughout both the "Practice" and "Challenge."
- **Expected From Students:**
 - Work individually or in small groups to design and build a robotic arm powered by syringes during the "Practice."
 - Apply their knowledge of hydraulic systems (from Pascal's Principle) to create movement and gripping functionality in the "Practice."
 - Select appropriate materials for construction in the "Practice."
 - Follow the provided instructions (or develop their own design) for the "Practice."
 - Troubleshoot any issues that arise during construction and operation in the "Practice."
 - Reflect on the strengths and limitations of their design in their responses to the reflection questions in the "Practice."
 - Engage with the "Challenge" by considering how to modify a robotic gripper to handle metallic items, proposing creative solutions based on their understanding of materials and forces.
- **Parts of the books included:**

Fill in the gaps to complete the robotic arm parts.



- **Activity - Practice & Challenge:**

- **Practice:**

- **Materials Needed:**

- Appropriate structural materials (e.g., cardboard, craft sticks, PVC pipes).
 - Syringes of various sizes.
 - Flexible tubing to connect the syringes.
 - Water to fill the hydraulic system.
 - Materials for creating a gripper (e.g., cardboard, wire, small clamps).
 - Fasteners (e.g., tape, glue, zip ties).

- **Step-by-Step Instructions:**

0. **Design:** Plan the structure of your robotic arm, including the base, arm segments, joints, and gripper. Consider how the

syringes and tubing will be used to create movement at the joints and control the gripper.

1. **Build the Structure:** Construct the physical framework of your robotic arm using the chosen materials. Ensure the joints are designed to allow for rotation or linear movement.
2. **Create Hydraulic Actuators:** Connect syringes with tubing, ensuring they are filled with water without air bubbles. These will act as your hydraulic cylinders.
3. **Connect Actuators to Joints:** Attach the syringes to the joints of your robotic arm in a way that pushing or pulling the syringe plungers will cause the arm segments to move.
4. **Design and Build Gripper:** Create a mechanism at the end of your arm that can grip small objects. Consider using a syringe to actuate the gripper (e.g., to open and close jaws).
5. **Test and Refine:** Experiment with your robotic arm, manipulating the syringes to make it move right, left, up, and down, and to grip objects. Identify any weaknesses in your design and make improvements.

- **Reflection Questions:**

0. How effectively did your hydraulic system allow the robotic arm to move and grip? What challenges did you encounter?
1. How did the size and arrangement of the syringes affect the force and range of motion of your arm?
2. What modifications could you make to your design to improve its strength, reach, or precision?
3. How does your model demonstrate Pascal's Principle in action?

- **Challenge:** Left for the students to be made at home.

- **Description (From the image):** Your robotic gripper is effective at holding various objects, but it struggles with metallic items. How could you modify the gripper's design to enhance its ability to attract

and hold onto metals? Consider what materials or components you could add to improve its functionality.

Workshop Document File:

<https://docs.google.com/document/d/1Q9CitJWeJp36tJghPtuDkznXyt3-p1-Y/edit?usp=sharing&ouid=112295837902879133584&rtpof=true&sd=true>



Time Frame: 90 mins

objectives:

- ☐ Use 3D models or diagrams to visualize how parts fit together.
- ☐ Create mechanical linkages to control the movement of a robotic arm.
- ☐ Solve problems by considering the 3D arrangement of components.

SEL Relation:

Challenge questions of the lesson:

1. Engage

- **Teacher Role:**
 - Pose intriguing questions about how complex machines move and are controlled.
 - Facilitate a brief introductory discussion on examples of mechanical linkages in everyday life and existing robotic systems.
 - Introduce the challenge of understanding and controlling movement in three-dimensional space.
 - Spark curiosity about how engineers design and visualize these intricate systems.
- **Expected From Students:**
 - Actively participate in the initial discussion, sharing examples of linkages they've encountered (e.g., bicycle brakes, door hinges, toys with moving parts).
 - Ask questions about the complexity of controlling movement in robots.
 - Show interest in the idea of visualizing and understanding 3D arrangements of mechanical components.
 - Begin to think about the relationship between different parts of a machine and how their connections dictate movement.

2. Think

- **Questions and Possible Answers:**
 - **Question 1 (From Image):** How does the placement of each part in a 3D space affect the balance and movement of the robotic arm? What might happen if a component is not positioned correctly?

- **Possible Answers:**

- Placement affects the center of gravity and overall balance; incorrect placement can lead to instability or tipping.
- Incorrect positioning of joints will limit the arm's range of motion or prevent intended movements.
- Misaligned linkages will not transfer motion correctly, leading to jerky or ineffective movement.
- Incorrect placement of actuators (motors, cylinders) will result in insufficient or misdirected force.
- Components might collide with each other if not positioned correctly in 3D space.

- **Question 2 (From Image):** How do 3D models help you see how the parts of the robotic arm fit together?

- **Possible Answers:**

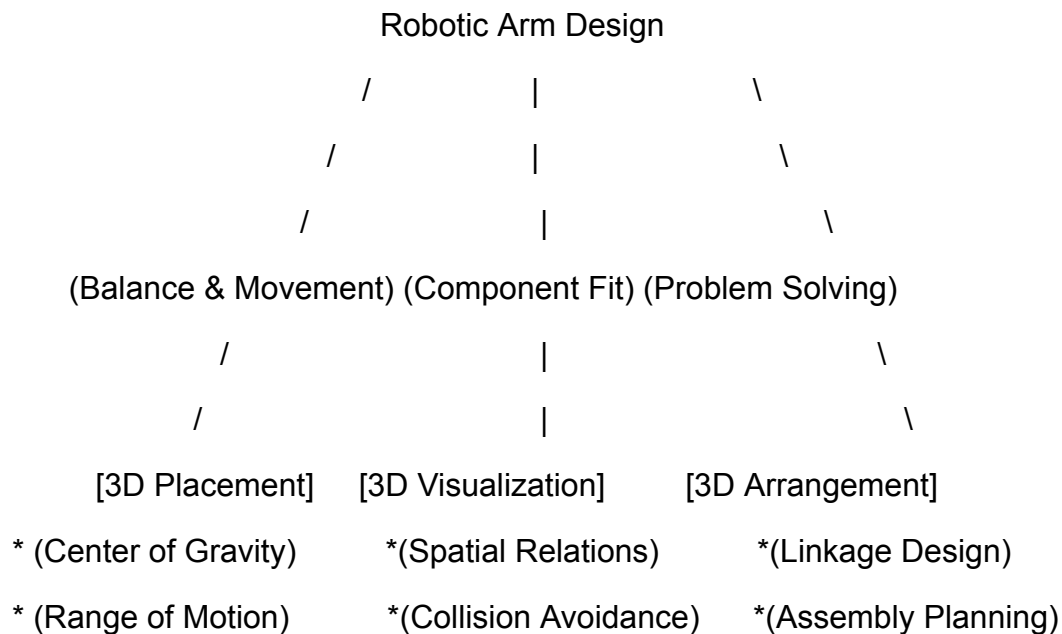
- They provide a visual representation of the spatial relationships between components from different angles.
- They allow you to see how parts will interact and move relative to each other in three dimensions.
- They can help identify potential clashes or interference between parts before physical construction.
- They can illustrate complex assemblies and how different linkages connect.
- They can make it easier to understand the overall structure and functionality of the arm.

- **Question 3 (From Image):** What challenges can you solve by visualizing the arm in 3D before building it?

- **Possible Answers:**

- Identifying and correcting design flaws related to movement range or collisions.
- Optimizing the placement of components for better balance and stability.
- Planning the assembly process and understanding the order in which parts need to be connected.
- Determining the necessary lengths and orientations of linkages.
- Visualizing the overall functionality and ensuring the design meets the intended purpose.
- Communicating design ideas more effectively with others.

Brainstorm Web:



(List strategies teachers can use to deliver this part, including the name of the strategy and how to implement it.)

- **Strategy 1: Think-Pair-Share:** Pose one question at a time, allowing individual thinking, partner discussion, and then whole-class sharing of ideas.
- **Strategy 2: "What If?" Scenarios:** Present hypothetical situations where a component is misplaced or a linkage is misaligned and ask students to predict the consequences.
- **Strategy 3: Using 3D Models (Physical or Digital):** Provide students with simple 3D models (even everyday objects connected in a way that mimics a jointed arm) and ask them to manipulate them to understand how placement affects movement and balance. For digital models, use rotation and zoom features to explore spatial relationships.
- **Strategy 4: Collaborative Diagramming:** Have small groups work together to sketch or create rough 3D diagrams of a robotic arm, focusing on how different parts connect and the potential challenges of incorrect placement.
- **Strategy 5: Role-Playing:** Assign students roles (e.g., a "balance engineer," a "motion control specialist," an "assembly planner") and have them discuss the importance of 3D arrangement from their specific perspective.

3.Explore

- **Teacher Role:**
 - Introduce Fusion 360 (or the chosen CAD software) and its basic interface relevant to creating 3D shapes.
 - Provide a brief overview of gears and their function in transmitting rotational motion.
 - Guide students on how to use the gear creation tools within Fusion 360.
 - Encourage experimentation with different gear parameters (number of teeth, size, etc.) to observe how they affect meshing.
 - Facilitate student exploration and problem-solving as they design gears that fit together.
- **Expected From Students:**

- Learn the basic navigation and tools within Fusion 360 necessary for creating and manipulating 3D objects.
 - Utilize the gear creation feature in Fusion 360 to generate different gears.
 - Experiment with various gear parameters to understand their impact on the gear's geometry.
 - Attempt to assemble two or more gears within Fusion 360 so that their teeth mesh correctly.
 - Observe and analyze how the 3D arrangement of the gears determines their ability to transmit motion.
- **Activity - Do It: Design gears that fit together in Fusion 360**
 - **Materials Needed:**
 - Computers with Fusion 360 (or alternative CAD software) installed.
 - Internet access (if Fusion 360 is cloud-based).
 - (Optional) Basic tutorial or handout on using the gear creation tool in Fusion 360.
 - **Step-by-Step Instructions (within Fusion 360):**
0. **Open Fusion 360:** Launch the software on your computer.
 1. **Create a New Design:** Start a new design file.
 2. **Access the Gear Tool:** Navigate to the "Create" menu. Look for options like "Gear," "Spur Gear," or similar. The exact location might vary slightly depending on the Fusion 360 version.
 3. **Define the First Gear:**
 - Select the gear creation tool.
 - You will likely be prompted to define parameters such as:
 - **Number of Teeth:** This determines the gear ratio and how it meshes with other gears. Start with a reasonable number (e.g., 20).

- **Pitch Diameter (or Module):** This controls the overall size of the gear and the size of its teeth. Keep a note of this value.
 - **Pressure Angle:** This is a standard value (often 20 degrees) that affects how the teeth engage.
 - **Thickness:** The width of the gear.
- Select a plane or origin point to place the gear.
 - Click "OK" to create the first gear.

4. **Create the Second Gear:**

- Repeat steps 3 and 4 to create a second gear.
- **Crucially, to make it mesh with the first gear, you will likely need to use the same Pitch Diameter (or Module) as the first gear.**
- You can experiment with a different **Number of Teeth** for the second gear to explore different gear ratios.

5. **Position the Gears:**

- Use the "Move/Copy" tool (found under the "Modify" menu or by right-clicking on a component).
- Select one of the gears and move it in the 3D workspace so that it is close to the first gear.
- Rotate one or both gears as needed to align their teeth visually.

6. **Observe the Meshing:**

- Zoom in closely to examine how the teeth of the two gears interact.

- Adjust the position and rotation of the gears until the teeth interlock without obvious gaps or overlaps. This demonstrates them "fitting together."

7. Experiment with Parameters (Optional but Recommended):

- Go back and modify the "Number of Teeth" of one or both gears. Observe how this affects the size and the way they need to be positioned to mesh.
- Try creating a third gear with a different number of teeth (but the same Pitch Diameter/Module) and attempt to mesh it with one of the existing gears.

○ **Reflection Questions:**

0. What parameters were important to consider when designing gears that needed to fit together?
1. How did changing the number of teeth on a gear affect its size and how it meshed with another gear?
2. How does the 3D arrangement (position and rotation) of the gears determine if they can successfully transmit motion?
3. What challenges did you encounter while trying to make the gears fit together in Fusion 360?
4. How might visualizing gears in 3D before physically building them be helpful in designing a robotic arm's mechanical linkages?

4. Explain

● **Teacher Role:**

- Facilitate a class discussion to connect the "Explore" activity (designing gears in Fusion 360) to the broader concepts of visualizing part fit, creating mechanical linkages, and considering 3D arrangements in robotic arms.
- Provide clear explanations of how CAD software like Fusion 360 is used by engineers to design and simulate mechanical systems.
- Highlight the importance of visualizing components in 3D to ensure proper fit and function before physical construction.

- Guide students to understand how gears and other mechanical linkages (which they will explore further) are fundamental for controlling movement in robotic arms.
- Use the "Watch It" resource to provide a visual demonstration of how Fusion 360 can be used to create robotic arm parts.
- **Expected From Students:**
 - Actively participate in the discussion, sharing their experiences and challenges from the gear design activity in Fusion 360.
 - Listen attentively to the teacher's explanation of the role of CAD software in engineering design.
 - Make connections between their 3D gear design experience and the idea of designing more complex robotic arm parts.
 - Formulate questions about how mechanical linkages and 3D arrangements are implemented in real robotic arms.
 - Engage with the "Watch It" video to see a practical demonstration of robotic arm part creation in Fusion 360.
- **Watch It:**



- **Pre-Watching: Set a purpose.**
 - **Teacher Role:** "Before we watch this video about using Fusion 360 to create robotic arm parts, think back to our gear design activity. What were some of the challenges in making the gears fit together correctly in 3D space? What do you hope to learn from this video about how engineers use software to design more complex parts for robotic arms and ensure they fit and function as intended?"
 - **Expected From Students:** Consider their experience with gear design in Fusion 360 and think about the complexities of designing entire robotic arm components. Formulate questions about the design process and how the software aids in visualization and fit.
- **During Watching: Active engagement.**
 - **Teacher Role:** "As you watch the video, pay attention to how the designer uses Fusion 360's tools to create the shapes of the robotic

arm parts. Notice how they might ensure that different parts will fit together correctly in a 3D assembly. Look for any techniques they use for visualizing the parts and their relationships."

- **Expected From Students:** Watch the video actively, observing the steps involved in creating robotic arm parts in Fusion 360, focusing on aspects related to shape creation, assembly considerations, and visualization techniques.
- **Post-Watching: Discussion and reflection.**
 - **Teacher Role:** "Now that we've watched the video, what are some of the key features of Fusion 360 that seem helpful for designing robotic arm parts? How does the software help engineers visualize how different components will fit together in 3D? What are some advantages of designing in a 3D environment like this before building physical prototypes? How does this relate to the challenges we considered when designing our gears?"
 - **Expected From Students:** Share their observations from the video, highlighting the capabilities of Fusion 360 for 3D design and visualization. Connect the software's features to the learning objectives and their own experiences with the gear design activity. Reflect on the benefits of using CAD software in the engineering design process.
 - **(List strategies teachers can use to deliver this part, including the name of the strategy and how to implement it.)**
 - **Strategy 1: Guided Note-Taking:** Provide students with a simple template with prompts to guide their viewing and note-taking during the video (e.g., "Software features I noticed," "How 3D visualization helps," "Connections to gear design").
 - **Strategy 2: "Pause and Discuss":** Pause the video at key points to ask students specific questions about what they have just seen and encourage immediate discussion.
 - **Strategy 3: Think-Write-Pair-Share:** After the video, have students individually write down their key takeaways, discuss with a partner, and then share with the class.

. Elaborate & Evaluate

- **Teacher Role:**

- Use the "Focus" assessment questions to gauge student understanding of key concepts related to robotic arm movement, design, and visualization in CAD software.
- Facilitate a discussion of the correct answers and the reasoning behind them, reinforcing the learning objectives.
- Introduce the "Practice" activity (likely building a physical linkage) as an opportunity for students to apply their knowledge in a hands-on way.
- Guide students during the "Practice," encouraging them to connect their physical construction to the 3D visualization concepts learned earlier.
- Observe and assess students' ability to create functional linkages and understand how they control movement.
- Encourage reflection on the relationship between 3D design, physical construction, and the control of robotic arm movement.
- **Expected From Students:**
 - Answer the "Focus" assessment questions, demonstrating their understanding of robotic arm components, gripper design considerations, and visualization tools in Fusion 360.
 - Participate in the discussion of the correct answers, explaining their reasoning and clarifying any misunderstandings.
 - Engage in the "Practice" activity by constructing a physical mechanical linkage.
 - Apply their understanding of how different parts connect and move in 3D space to build a functional linkage.
 - Observe how the physical linkage translates motion and relates to the concepts explored in Fusion 360.
 - Reflect on the challenges and insights gained from the physical construction process.
- **Assessment - Focus:**

Assessment

Focus



1. Which of the following components is crucial for allowing movement at different angles in a robotic arm?

- A) Base.
- B) End Effector.
- C) Joint.
- D) Actuator.

2. When designing a gripper for a robotic arm, which feature is most important to ensure it can securely hold various objects?

- A) The color of the gripper.
- B) The size and shape of the gripper pads.
- C) The weight of the gripper.
- D) The material used for the base of the robotic arm.

3. In Fusion 360, which tool would you use to test how the robotic arm's parts interact and move together?

- A) Sketch Tool.
- B) Extrude Tool.
- C) Simulation Tool.
- D) Render Tool.

Practice:

- **Activity:** Design the robotic arm body in Fusion 360.
- **Materials Needed:** Computers with Fusion 360 (or alternative CAD software) installed, internet access (if cloud-based), potentially tutorials or guides on basic 3D modeling in Fusion 360.
- **Step-by-Step Instructions (within Fusion 360):**
 1. **Open Fusion 360 and Create a New Design.**
 2. **Sketch the Base:** Use the Sketch tools to create a base for the robotic arm. Consider its stability and how it might be mounted.
 3. **Create the First Arm Segment:** Sketch the profile of the first part of the arm. Use the Extrude tool to give it 3D thickness.

4. **Model Joint Locations:** Think about where the joints will be located to allow for movement. You can represent these as features or plan for them in the design of the arm segments.
 5. **Create Subsequent Arm Segments:** Design and extrude additional segments of the arm, considering how they will connect to the previous segments at the planned joint locations.
 6. **Visualize Assembly:** Use the Assembly features in Fusion 360 to move the different segments relative to each other, imagining how the joints will function. (Note: Full assembly with constraints might come later, but basic movement can be visualized).
 7. **Save Your Design.**
- **Reflection Questions:**
 - What considerations did you have when designing the different segments of the robotic arm body in 3D space?
 - How did you plan for the placement of joints to allow for a useful range of motion?
 - What challenges did you encounter while creating the 3D model of the arm body?
 - How does designing the arm body in Fusion 360 help you visualize how the different parts will fit together in a physical robot?

for Video Explanations:

<https://drive.google.com/file/d/1YvujceboyueGM4XtDid4Pa4XwJPSySFA/view>